

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID - SEM EXAMINATION, SUMMER, 2025
SESSION: 2024 – 2025 (Spring)
Guided-Self

Subject code: CS6510

Subject Name: Deep Learning

Dept. Code: CS

No. of pages: 1

Full Marks: 30

Duration: 2 Hrs

Answer ALL questions. Unnecessary and extra writing will attract negative marks.

Q. No.	Questions	Marks	CO	BL
1.	(a) Explain the architecture of GRU. Explain the vanishing gradient and exploding gradient problems. (b) What is the Softmax activation function? How is it different from the Sigmoid function, and when would Softmax be used instead of Sigmoid in a neural network?	6+4	CO1	BL1
2.	(a) Explain the Adam optimization algorithm. What are its key components, and how does it differ from standard stochastic gradient descent? What is the advantage of using momentum (first moment estimate) in gradient-based optimization? (b) Discuss the Overfitting and Underfitting problems. Explain L1 and L2 regularization.	6+4	CO2	BL2
3.	(a) A CNN applies a 3×3 convolution kernel to a 6×6 input image with padding = 1 (stride = 1). Calculate the output dimensions after the convolution. What will be the final output dimensions if a 2×2 max pooling layer (stride = 2) is applied to the result? (b) In a ResNet bottleneck block, the input has 256 channels. The block uses a 1×1 convolution to reduce dimensions to 64 channels, followed by a 3×3 convolution with 64 filters, and another 1×1 convolution to restore dimensions to 256 channels. Assume the input feature map size is 56×56. (i) What are the output dimensions (height × width × channels) after each convolutional layer in the bottleneck block (assume padding and stride = 1 for all layers)? (ii) Why is the use of 1×1 convolution beneficial in such bottleneck architectures?	5+5	CO3	BL3

CO: Course Outcome and BL: Bloom's Level

***** ALL THE BEST *****